Course Outline
Week 1
Orientation & Embedded System Architecuture

***Orientation* Objectives**

- Introduction to the course syllabus
- Post self-introduction message to a discussion forum

Things to Do:

- Order the development kit
- Obtain, install and use the development software tools
- Obtain the required textbook

- ***Lesson Learning Objectives:***
- Define what distinguishes an ‘Embedded System' from other computer systems

Assignments

- Read Section 1.2 of the text book (Computers as Components)
- Read the setup instructions for the software tools and the Development kit
- LAB#1

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Course Outline
Week 2
Embedded System Design Process & COTS Modules

***Learning* Objectives:**

- Describe the importance of an Embedded System Abstraction
- Identify the key components required in an Embedded System Design
- Evaluating Commercial-Off-The-Shelf modules
- BOM generation and hardware selection strategies

Assignments

- Read Section 1.3 of the text book (Computers as Components)
- Quiz #1
- LAB#2

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Course Outline
Week 3
Selecting Microcontrollers

***Learning* Objectives:**

- Computer Architecture Basics
- Compare the different types of processor architectures
- ARM Cortex-M architecture

Assignments

- Read section 2.2, 2.3, 3.2, and 3.3 from textbook
- Quiz #2

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Course Outline
Week 4
Caches and Memory Management Units(MMU)

***Learning* Objectives:**

- Identify different forms of Cache algorithms
- Discuss the advantages of having an MMU
- Types of embedded memory

Assignments

- Read section 3.5, 3.6 from textbook
- Quiz #3
- LAB#3

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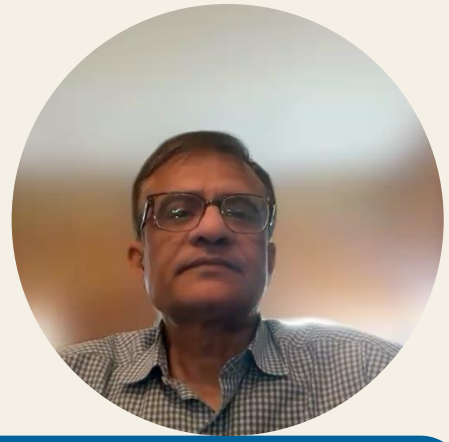
Course Outline
Week 5
Bus based CPU Systems and Sensors

***Learning* Objectives:**

- Design and develop using CPU-based systems
- Sensor categories and characteristics
- Reading sensor datasheets

Assignments

- Read sections 4.2,4.3,4.4, and 4.8 from the textbook
- Quiz #4
- LAB#4



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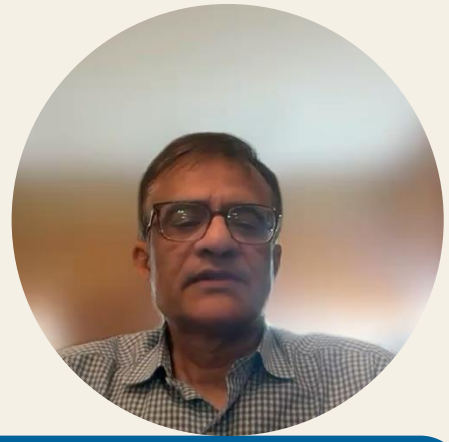
Course Outline
Week 6
Internet of Things and Automotive Systems

***Learning* Objectives:**

- Internal buses: AHB, APB, DMA
- External interfaces: I^2C, SPI, UART, USB, CAN
- Wireless interface overview (BLE, Wi-Fi, Sub-GHz)
- Interface selection based on bandwidth, latency, and power

Assignments

- Read sections 8.2,8.3,8.4,9.2,9.3,9.4 from the textbook
- Quiz #5
- LAB#5



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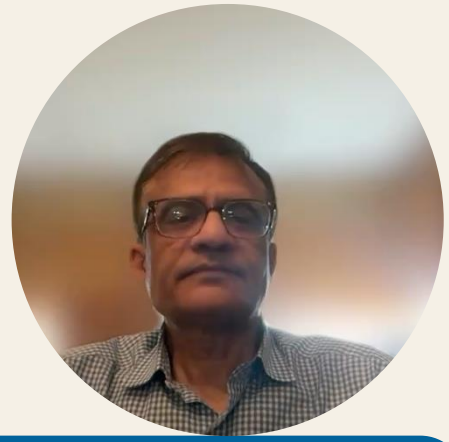
Course Outline
Week 7
PCB Design Foundations and Hardware Integration

***Learning* Objectives:**

- PCB stackups, routing rules, and decoupling
- Power integrity and grounding techniques
- High-speed design considerations
- Understanding eval board PCBs using the STM32 kit as a reference

Assignments

- Read the material provided in the module
- Quiz #6
- LAB#6



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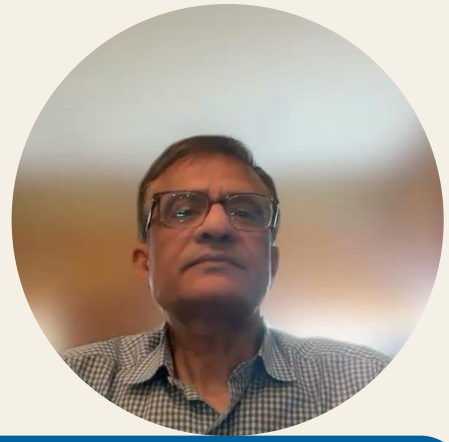
Course Outline
Week 8
System Design Techniques

***Learning* Objectives:**

- How proper design methodology can speed up time to market and improve quality & cost
- Identify various Design Methodologies and understand which suits best for your needs
- Explain the benefits of having good Requirements and Specifications
- Explain why Quality needs to be built into the product

Assignments

- Read the material provided in the module
- Quiz #7



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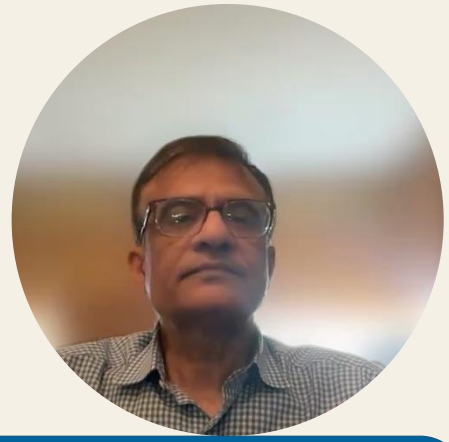
Course Outline
Week 9
Capstone: Build & Test a Complete Embedded System

***Learning* Objectives:**

- System integration strategies
- Power and performance optimization
- Preparing engineering documentation
- Common hardware/firmware integration pitfalls

Assignments

- Read the material provided in the module
- Final project details provided in the module



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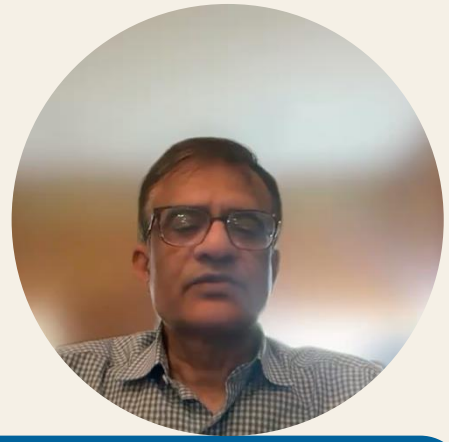
Course Outline

Evaluation of Student Performance

Discussion Board participation:	10%
Lab Assignments	40%
Quizzes	30%
Final project.	20%
TOTAL	100%



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